Conditional Probability and Bayes Rule

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises**

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.05$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

Final Answer: 0.0450. By the multiplication rule of conditional probability:

$P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.05 \times 0.90 = 0.0450$.

2 **[Foundational]** • **Target: Joint Probability and Independence**

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.10$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

Final Answer: 0.0900. By the multiplication rule of conditional probability:

$P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.10 \times 0.90 = 0.0900$.

3 **[Foundational]** • **Target: Conditional Probability and the Chain Rule**

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.15$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

Final Answer: 0.1350. By the multiplication rule of conditional probability:

$P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.15 \times 0.90 = 0.1350$.

4 **[Foundational]** • **Target: Law of Total Probability**

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.20$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

Final Answer: 0.1800. By the multiplication rule of conditional probability:

$P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.20 \times 0.90 = 0.1800$.

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5 [Foundational] • Target: Bayes Theorem in Machine Learning and Screening

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.25$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

Final Answer: 0.2250. By the multiplication rule of conditional probability:

$$P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.25 \times 0.90 = 0.2250.$$

6 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.30$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

Final Answer: 0.2700. By the multiplication rule of conditional probability:

$$P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.30 \times 0.90 = 0.2700.$$

7 [Foundational] • Target: Joint Probability and Independence

In anomaly detection, anomaly prior $P(A) = 0.02$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly? Show your complete algebraic work.

Final Answer: 0.279. By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A) + P(+|egA)P(egA)} = \frac{0.95 \times 0.02}{0.95(0.02) + 0.05(0.98)} = \frac{0.0190}{0.0680} = 0.279.$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Conditional Probability and the Chain Rule

In anomaly detection, anomaly prior $P(A) = 0.04$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly? Show your complete algebraic work.

Final Answer: 0.442. By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A) + P(+|egA)P(egA)} = \frac{0.95 \times 0.04}{0.95(0.04) + 0.05(0.96)} = \frac{0.0380}{0.0860} = 0.442.$

9 [Intermediate] • Target: Law of Total Probability

In anomaly detection, anomaly prior $P(A) = 0.06$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly? Show your complete algebraic work.

Final Answer: 0.548. By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A) + P(+|egA)P(egA)} = \frac{0.95 \times 0.06}{0.95(0.06) + 0.05(0.94)} = \frac{0.0570}{0.1040} = 0.548.$

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10 **[Intermediate]** • **Target: Bayes Theorem in Machine Learning and Screening**

In anomaly detection, anomaly prior $P(A) = 0.08$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly? Show your complete algebraic work.

Final Answer: 0.623. By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A) + P(+|egA)P(egA)} = \frac{0.95 \times 0.08}{0.95(0.08) + 0.05(0.92)} = \frac{0.0760}{0.1220} = 0.623$.

11 **[Intermediate]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises**

In anomaly detection, anomaly prior $P(A) = 0.10$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly? Show your complete algebraic work.

Final Answer: 0.679. By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A) + P(+|egA)P(egA)} = \frac{0.95 \times 0.10}{0.95(0.10) + 0.05(0.90)} = \frac{0.0950}{0.1400} = 0.679$.

12 **[Intermediate]** • **Target: Joint Probability and Independence**

In anomaly detection, anomaly prior $P(A) = 0.12$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly? Show your complete algebraic work.

Final Answer: 0.722. By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A) + P(+|egA)P(egA)} = \frac{0.95 \times 0.12}{0.95(0.12) + 0.05(0.88)} = \frac{0.1140}{0.1580} = 0.722$.

13 **[Intermediate]** • **Target: Conditional Probability and the Chain Rule**

Consider a medical diagnostic test for a rare disease. The prevalence of the disease in the general population is $P(D) = 0.001$. The test has a sensitivity of $P(+|D) = 0.99$ (the probability of a positive test given the person has the disease) and a specificity of $P(-|egD) = 0.98$ (the probability of a negative test given the person does not have the disease). If a randomly selected individual tests positive, what is the probability that they actually have the disease, given that we are interested in the positive predictive value (PPV) of the test, and the prior probability of having the disease is adjusted by a factor f such that the new prior is $P'(D) = f \cdot P(D)$? For this specific scenario, suppose $f = 1.5$ and we want to find $P'(D|+)$.

Final Answer: $\frac{0.99 \cdot 0.0015}{0.99 \cdot 0.0015 + 0.02 \cdot (1 - 0.0015)}$. Let D be the event that an individual has the disease, and + be the event that the test is positive. We are given: - Prior probability of disease: $P(D) = 0.001$ - Sensitivity: $P(+|D) = 0.99$ - Specificity: $P(-|egD) = 0.98$ From specificity, we can find the false positive rate: $P(+|egD) = 1 - P(-|egD) = 1 - 0.98 = 0.02$. We are considering a scenario where the prior probability is adjusted by a factor.

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14 **[Intermediate]** • **Target: Law of Total Probability**

A quality control engineer is monitoring the number of defects produced by two different machines, Machine A and Machine B. Let X be the number of defects produced by Machine A in a given hour, and Y be the number of defects produced by Machine B in the same hour. The joint probability distribution of X and Y is given in the table below: | $P(X=x, Y=y)$ | $y=0$ | $y=1$ | $y=2$ | | :---------------: | :-------: | :-------: | :-------: | | $x=0$ | 0.10 | 0.15 | 0.05 | | $x=1$ | 0.20 | 0.30 | 0.10 | | $x=2$ | 0.05 | 0.05 | 0.00 | What is the expected value of the total number of defects produced by both machines in a given hour, $E(X+Y)$? Show your complete algebraic work.

Final Answer: 1.30. To find the expected value of the total number of defects, $E(X+Y)$, we can use the property of linearity of expectation: $E(X+Y) = E(X) + E(Y)$. First, let's calculate the marginal probability distributions for X and Y. For X: $P(X=0) = 0.10 + 0.15 + 0.05 = 0.30$
$P(X=1) = 0.20 + 0.30 + 0.10 = 0.60$ $P(X=2) = 0.05 + 0.05 + 0.00 = 0.10$ Now, calculate $E(X)$:
$E(X) = (0 \times 0.30) + (1 \times 0.60) + (2 \times 0.10) = 0 + 0.60 + 0.20 = 0.80$ For Y: $P(Y=0) = 0.10 + 0.20 + 0.05 = 0.35$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Bayes Theorem in Machine Learning and Screening** • **Cross-Lesson: 5.01 Distributions and Maximum Likelihood**

A medical diagnostic test for a rare disease is designed to be highly sensitive and specific. The disease affects 1 in every 1000 individuals in the population. The test has a sensitivity of 99% (meaning it correctly identifies 99% of individuals who have the disease) and a specificity of 95% (meaning it correctly identifies 95% of individuals who do not have the disease). If a randomly selected individual from the population tests positive, what is the probability that they actually have the disease? Show your complete algebraic work.

Final Answer: Approximately 0.019. Let D be the event that an individual has the disease, and P be the event that the individual tests positive. We are given the following probabilities: - Prior probability of having the disease: $P(D) = \frac{1}{1000} = 0.001$ - Prior probability of not having the disease:
$P(D^c) = 1 - P(D) = 1 - 0.001 = 0.999$ - Sensitivity (true positive rate): $P(P|D) = 0.99$ - Specificity (true negative rate): $P(P^c|D^c) = 0.95$ From specificity, we can derive the false positive rate.

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16 **[Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS**

A medical test for a rare disease has a 99\% accuracy rate in detecting the disease when it is present (True Positive Rate) and a 95\% accuracy rate in identifying individuals without the disease (True Negative Rate). If the prevalence of the disease in the population is 0.5\%, what is the probability that a person who tests positive actually has the disease? Show your complete algebraic work.

Final Answer: Approximately 9.1\%. Let D be the event that a person has the disease, and P be the event that the person tests positive. We are given: P(P|D) = 0.99 (True Positive Rate) P(not P|not D) = 0.95 (True Negative Rate), which means P(P|not D) = 1 - 0.95 = 0.05 (False Positive Rate) P(D) = 0.005 (Prevalence of the disease) From P(D), we know P(not D) = 1 - P(D) = 1 - 0.005 = 0.995. We want to find P(D|P), the probability that a person has the disease given a positive test result. Using Bayes' Theorem: P(D|P.

17 **[Challenging] • Target: Joint Probability and Independence • Cross-Lesson: 7.01 Logistic Regression and Classification**

Consider two events, A and B, with the following probabilities: $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. What is the conditional probability of A given B, denoted as $P(A|B)$? Show your complete algebraic work.

Final Answer: 0.4. The formula for conditional probability is $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Given $P(A \cap B) = 0.2$ and $P(B) = 0.5$, we can substitute these values into the formula: $P(A|B) = \frac{0.2}{0.5} = 0.4$. Therefore, the conditional probability of A given B is 0.4.

18 **[Challenging] • Target: Conditional Probability and the Chain Rule • Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.10$. What is the probability of the complement class $P(egC_1)$? Show your complete algebraic work.

Final Answer: 0.90. By the complement rule of probability: $P(egC_1) = 1 - P(C_1) = 1 - 0.10 = 0.90$.

19 **[Challenging] • Target: Law of Total Probability • Cross-Lesson: 1.01 Vectors and Embeddings in AI**

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.20$. What is the probability of the complement class $P(egC_1)$? Show your complete algebraic work.

Final Answer: 0.80. By the complement rule of probability: $P(egC_1) = 1 - P(C_1) = 1 - 0.20 = 0.80$.

20 **[Challenging] • Target: Bayes Theorem in Machine Learning and Screening • Cross-Lesson: 2.01 Neural Layer Matrices and Activations**

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.60$. What is the probability of the complement class $P(egC_1)$? Show your complete algebraic work.

Final Answer: 0.40. By the complement rule of probability: $P(egC_1) = 1 - P(C_1) = 1 - 0.60 = 0.40$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.10$. What is the probability of the complement class $P(\neg C_1)$?

a. **0.90** ✓ **b.** 0.95

c. 0.05 **d.** 1.10

By the complement rule of probability: $P(\neg C_1) = 1 - P(C_1) = 1 - 0.10 = 0.90$.

22 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.20$. What is the probability of the complement class $P(\neg C_1)$?

a. **0.80** ✓ **b.** 0.90

c. 0.10 **d.** 1.20

By the complement rule of probability: $P(\neg C_1) = 1 - P(C_1) = 1 - 0.20 = 0.80$.

23 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.30$. What is the probability of the complement class $P(\neg C_1)$?

a. **0.70** ✓ **b.** 0.85

c. 0.15 **d.** 1.30

By the complement rule of probability: $P(\neg C_1) = 1 - P(C_1) = 1 - 0.30 = 0.70$.

24 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.40$. What is the probability of the complement class $P(\neg C_1)$?

a. **0.60** ✓ **b.** 0.80

c. 0.20 **d.** 1.40

By the complement rule of probability: $P(\neg C_1) = 1 - P(C_1) = 1 - 0.40 = 0.60$.

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25 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.50$. What is the probability of the complement class $P(\neg C_1)$?

a. **0.50** ✓

b. 0.75

c. 0.25

d. 1.50

By the complement rule of probability: $P(\neg C_1) = 1 - P(C_1) = 1 - 0.50 = 0.50$.

26 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.60$. What is the probability of the complement class $P(\neg C_1)$?

a. **0.40** ✓

b. 0.70

c. 0.30

d. 1.60

By the complement rule of probability: $P(\neg C_1) = 1 - P(C_1) = 1 - 0.60 = 0.40$.

27 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A machine learning diagnostic test has sensitivity $P(\text{Positive} | \text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.05$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

a. **0.0450** ✓

b. 0.9500

c. 0.8500

d. 0.0050

By the multiplication rule of conditional probability: $P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive} | \text{Defect}) = 0.05 \times 0.90 = 0.0450$.

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28 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A machine learning diagnostic test has sensitivity \(P(\text{Positive} | \text{Defect}) = 0.90 \). If the defect rate in production is \(P(\text{Defect}) = 0.10 \), find the joint probability of a defect and a positive test result \(P(\text{Defect} \cap \text{Positive}) \).

a. **\(0.0900 \)** ✓

b. \(1.0000 \)

c. \(0.8000 \)

d. \(0.0100 \)

By the multiplication rule of conditional probability: \(P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive} | \text{Defect}) = 0.10 \times 0.90 = 0.0900 \).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Joint Probability and Independence

If two events A and B are statistically independent with $P(A) = 0.4$ and $P(B) = 0.3$, find the joint probability $P(A \cap B)$.

a. 0.12 ✓

b. 0.70

c. 0.10

d. 0.30

By independence: $P(A \cap B) = P(A) \times P(B) = 0.4 \times 0.3 = 0.12$.

30 [Intermediate] • Target: Joint Probability and Independence

Given $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cup B) = 0.7$, find $P(A \cap B)$.

a. 0.20 ✓

b. 0.10

c. 0.30

d. 0.15

$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.5 + 0.4 - 0.7 = 0.20$.

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31 [Intermediate] • Target: Joint Probability and Independence

Are the events in question 2 statistically independent?

a. **Yes, because** $P(A)P(B) = 0.5 \times 0.4 = 0.20 = P(A \cap B)$ ✓

b. No, because $P(A) \neq P(B)$

c. No, because $P(A \cup B) \neq 1$

d. Cannot be determined without conditional probabilities

Since $P(A \cap B) = 0.20 = P(A)P(B)$, the definition of independence is satisfied exactly.

32 [Intermediate] • Target: Conditional Probability and the Chain Rule

Given $P(A) = 0.6$ and $P(A \cap B) = 0.18$, calculate the conditional probability $P(B|A)$.

a. 0.30 ✓

b. 0.108

c. 0.42

d. 0.25

$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.18}{0.60} = 0.30$.

33 [Intermediate] • Target: Conditional Probability and the Chain Rule

In an autoregressive language model, write the joint probability of a sequence of three tokens $P(w_1, w_2, w_3)$ using the probability Chain Rule.

a. $P(w_1)P(w_2|w_1)P(w_3|w_1, w_2)$ ✓

b. $P(w_1) + P(w_2) + P(w_3)$

c. $P(w_1)P(w_2)P(w_3)$

d. $P(w_3|w_2)P(w_2|w_1)$

By exact probability decomposition: $P(w_1, w_2, w_3) = P(w_1)P(w_2|w_1)P(w_3|w_1, w_2)$.

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34 [Intermediate] • Target: Conditional Probability and the Chain Rule

If $P(A|B) = 0.8$ and $P(B) = 0.25$, find $P(A \cap B)$.

a. 0.20 ✓

b. 0.55

c. 0.3125

d. 0.10

$P(A \cap B) = P(A|B)P(B) = 0.8 \times 0.25 = 0.20$.

35 [Intermediate] • Target: Law of Total Probability

Factory line 1 produces 60% of microchips with defect rate 2%. Line 2 produces 40% with defect rate 5%. What is the total probability that a random chip is defective?

a. 0.032 ✓

b. 0.070

c. 0.035

d. 0.025

$P(D) = P(D|L_1)P(L_1) + P(D|L_2)P(L_2) = (0.02)(0.60) + (0.05)(0.40) = 0.012 + 0.020 = 0.032$.

36 [Intermediate] • Target: Law of Total Probability

In question 7, if a selected chip is defective, what is the probability it came from Line 2?

a. $\frac{0.020}{0.032} = 0.625$ ✓

b. 0.400

c. 0.375

d. 0.500

Bayes: $P(L_2|D) = \frac{P(D|L_2)P(L_2)}{P(D)} = \frac{0.020}{0.032} = 0.625$.

37 [Intermediate] • Target: Law of Total Probability

State the Law of Total Probability for an exhaustive partition $\{A_1, A_2, \ldots, A_k\}$ of the sample space.

a. $P(B) = \sum_{i=1}^{k} P(B|A_i)P(A_i)$ ✓

b. $P(B) = \prod_{i=1}^{k} P(B|A_i)$

c. $P(B) = \max_i P(B|A_i)$

d. $P(B) = \sum_{i=1}^{k} P(A_i|B)$

Every element in B belongs to exactly one partition A_i, giving $P(B) = \sum P(B \cap A_i) = \sum P(B|A_i)P(A_i)$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Bayes Theorem in Machine Learning and Screening • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

In medical screening, a disease has prevalence $P(D) = 0.01$. The test has sensitivity $P(+|D) = 0.99$ and false positive rate $P(+|\sim D) = 0.02$. What is the posterior probability $P(D|+)$ that a positive patient has the disease?

a. $\approx 33.3\%$ ✓

b. 99.0%

c. 50.0%

d. 5.0%

$P(+) = (0.99)(0.01) + (0.02)(0.99) = 0.0099 + 0.0198 = 0.0297$. $P(D|+) = \frac{0.0099}{0.0297} = \frac{1}{3} \approx 33.3\%$.

39 [Challenging] • Target: Bayes Theorem in Machine Learning and Screening • Cross-Lesson: 1.01 Vectors and Embeddings in AI

What is the foundational simplifying assumption of the Naive Bayes text classifier?

a. **Features (words) are conditionally independent given the class label** ✓

b. All prior probabilities are equal

c. The vocabulary size is infinite

d. Features follow a uniform distribution

Naive Bayes assumes $P(x_1, \ldots, x_n|C) = \prod_{i=1}^{n} P(x_i|C)$, dramatically reducing parameter count.

40 [Challenging] • Target: Bayes Theorem in Machine Learning and Screening • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

In Maximum A Posteriori (MAP) estimation, how does the optimization objective differ from standard Maximum Likelihood Estimation (MLE)?

a. **MAP incorporates a prior distribution:** $\theta_{MAP} = \arg\max[\ln P(D|\theta) + \ln P(\theta)]$ ✓

b. MAP ignores all observed training data

c. MAP only applies to linear models

d. MAP minimizes variance instead of error

MAP maximizes posterior $\propto$ likelihood $\times$ prior, functioning mathematically like regularized MLE.

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41 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Consider a dataset of n data points, each residing in $\mathbb{R}^D$. We are performing Principal Component Analysis (PCA) to reduce the dimensionality. The first three principal components capture 70% of the total variance, the first six capture 85%, and the first ten capture 92%. You are tasked with selecting a subset of principal components for a downstream machine learning model. If the model requires that at least 90% of the original variance be retained, and you want to use the minimum possible number of principal components, what is the dimensionality of the transformed data?

a. 3

b. 6

c. **10** ✓

d. D

The problem requires us to find the minimum number of principal components such that the cumulative explained variance is at least 90%. The provided information states: - The first 3 principal components capture 70% of the variance. - The first 6 principal components capture 85% of the variance. - The first 10 principal components capture 92% of the variance. We need to find the smallest k such that the cumulative variance captured by the first k principal components is $\geq 90\%$. Checking the giv.

42 [Challenging] • Target: Joint Probability and Independence • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Consider a discrete random variable X with probability mass function (PMF) given by $P(X = k; \theta) = \frac{\theta^k e^{-\theta}}{k!}$ for $k = 0, 1, 2, \ldots$, where $\theta > 0$ is a parameter. Suppose we have an i.i.d. sample $x_1, x_2, \ldots, x_n$ from this distribution. We want to find the maximum likelihood estimate (MLE) for θ that maximizes the expected value of θ conditional on the sum of the observations, $S = \sum_{i=1}^{n} x_i$. Specifically, we are interested in the value of θ that maximizes $E[\theta | S = s]$. For a Poisson distribution, it is known that $E[\theta | S = s] = \frac{\theta s}{\theta + s}$ if θ itself were distributed as $Gamma(s + 1, 1)$. However, here θ is a fixed but unknown parameter. This question is about finding a parameter value that optimizes a related quantity. Let's consider a modified objective function for estimation that relates to the Fisher Information. Suppose we are interested in maximizing $L(\theta) = \frac{1}{n} \sum_{i=1}^{n} \log P(x_i; \theta) - \theta$. Find the value of θ that maximizes this modified likelihood function, given a sample where $n = 5$ and $\sum_{i=1}^{5} x_i = 15$.

a. 3 ✓

b. 5

c. 15

d. 2

The given distribution is a Poisson distribution with parameter θ, so $P(X = k; \theta) = \frac{\theta^k e^{-\theta}}{k!}$. The log-likelihood for a single observation x_i is $\log P(x_i; \theta) = x_i \log \theta - \theta - \log(x_i!)$. The log-likelihood for the entire sample is $\log L(\theta; x_1, \ldots, x_n) = \sum_{i=1}^{n} (x_i \log \theta - \theta - \log(x_i!)) = \log \theta \sum_{i=1}^{n} x_i - n\theta - \sum_{i=1}^{n} \log(x_i!)$. Let $S = \sum_{i=1}^{n} x_i$. Then.

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43 [Challenging] • Target: Conditional Probability and the Chain Rule • Cross-Lesson: 6.01 Linear Regression and OLS

A national health organization is investigating the relationship between daily screen time (in hours) and the number of hours of sleep reported by adults. They collect data from a sample of 1200 adults. The data is summarized in a two-way table, where the row totals represent categories of daily screen time and the column totals represent categories of reported sleep hours. The organization is particularly interested in the cells where adults report less than 4 hours of screen time and get 7 or more hours of sleep. If the overall row total for 'less than 4 hours of screen time' is 250, and the overall column total for '7 or more hours of sleep' is 300, what is the expected count for the cell representing adults with less than 4 hours of screen time who get 7 or more hours of sleep, assuming independence between screen time and sleep hours?

a. 62.50 ✓

b. 75.00

c. 83.33

d. 250.00

The problem asks for the expected count in a specific cell of a contingency table, assuming independence between the two categorical variables: daily screen time and reported sleep hours. The formula for the expected count in a cell is given by: $E = \frac{\text{(Row Total)} \times \text{(Column Total)}}{\text{Grand Total}}$ In this scenario: - The Grand Total sample size is given as 1200 adults. - The Row Total for the category 'less than 4 hours of screen time' is 250. - The Column Total for the category.

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44 [Challenging] • Target: Law of Total Probability • Cross-Lesson: 7.01 Logistic Regression and Classification

A machine dispenses a liquid into containers. The amount of liquid dispensed, in milliliters, follows a continuous uniform distribution over the interval $[95, 105]$. A quality control inspector randomly selects 5 containers and measures the amount of liquid in each. Let X_i be the amount of liquid in the ith container, for $i = 1, 2, 3, 4, 5$. The company has implemented a new calibration process that, on average, increases the mean amount of liquid dispensed by δ milliliters, but also increases the variance by σ^2. The inspector is specifically looking for evidence that the calibration process has *decreased* the variability of the dispensed liquid while maintaining a satisfactory average fill. If the current calibration process yields a mean fill of $\mu_0 = 100$ ml and a standard deviation of $\sigma_0 = 2$ ml, and the new process is hypothesized to have a mean fill of $\mu_1 = 100 + \delta$ ml and a standard deviation of $\sigma_1 = 2 - \sigma_{dev}$ ml, where $\delta > 0$ and $\sigma_{dev} > 0$ are unknown positive constants, what is the minimum value of σ_{dev} such that the probability of falsely concluding that the new process has *increased* variability (i.e., Type I error) is at most 0.01, assuming the mean fill remains at 100 ml (i.e., $\delta = 0$)?

a. The probability cannot be determined without knowing the specific value of δ.

b. Approximately 1.01

c. **Approximately** 1.37 ✓

d. Approximately 2.00

Let X be the amount of liquid dispensed by the machine. X follows a continuous uniform distribution $U(95, 105)$. The mean is $\mu = \frac{95+105}{2} = 100$ ml. The variance is $\sigma^2 = \frac{(105-95)^2}{12} = \frac{100}{12} = \frac{25}{3}$. The standard deviation is $\sigma = \sqrt{\frac{25}{3}} = \frac{5}{\sqrt{3}} \approx 2.887$ ml. The problem statement implies the initial standard deviation is $\sigma_0 = 2$ ml, which contradicts the calculated standard deviation for a uniform distribution from 95 to 10.

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45 [Challenging] • Target: Bayes Theorem in Machine Learning and Screening • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

A manufacturing company is testing a new quality control process. They have collected data on the number of defects found in batches of a product, categorized by the production shift. The data is summarized in the two-way table below: | Shift | Defects Found | No Defects Found | Total | |---|---|---|---| | Morning | 15 | 85 | 100 | | Afternoon | 25 | 75 | 100 | | Evening | 10 | 90 | 100 | | **Total** | **50** | **250** | **300** | If a test of independence is to be performed on this data, what is the expected number of defects found in the afternoon shift?

a. 20.00

b. **16.67** ✓

c. 25.00

d. 8.33

To find the expected number of defects found in the afternoon shift for a test of independence, we use the formula for expected counts: $E_{ij} = \frac{\text{(Row Total)} \times \text{(Column Total)}}{\text{Grand Total}}$ In this case, for the cell representing "Afternoon" shift and "Defects Found": - Row Total (Afternoon) = 100 - Column Total (Defects Found) = 50 - Grand Total = 300 Therefore, the expected count is.

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